ENGINEERING GEOLOGY

UNIT - II : MINERALOGY AND PETROLOGY

Minerals, Rocks - Igneous, Sedimentary & Metamorphic

Syllabus (as given)

- Definitions of mineral and rock - methods of study. Physical properties of minerals and rocks (megascopic). Rock forming minerals: Feldspar, Quartz, Olivine, Augite, Hornblende, Mica, Asbestos, Talc, Chlorite, Kyanite, Garnet, Calcite; ore minerals: Pyrite, Hematite, Magnetite, Galena, Pyrolusite, Graphite, Chromite, Bauxite. Classification, structures, textures and forms of Igneous, Sedimentary and Metamorphic rocks; megascopic study of granite varieties (pink, gray, green), Pegmatite, Dolerite, Basalt, Shale, Sandstone, Limestone, Laterite, Quartzite, Gneiss, Schist, Marble, Khondalite and Slate.

1. Mineral and Rock - Definitions

- **Mineral**: a naturally occurring, **inorganic, homogeneous solid** with a definite chemical composition (or narrow range) and an ordered **crystalline atomic structure**. Eg: quartz (SiO_2), calcite ($CaCO_3$).
- **Rock**: a natural **aggregate of one or more minerals** (or mineral matter) forming a part of the earth's crust. Eg: granite = quartz + feldspar + mica.
  - Monomineralic rock: limestone (calcite), quartzite (quartz), marble. Polymineralic: granite, gneiss.
- **Crystal**: a mineral with regular geometric shape bounded by plane faces; **Amorphous**: no regular structure (opal, volcanic glass).
- About **2 – 3 thousand minerals** are known, but only ~**12 rock-forming minerals** make up 99% of the crust; **Silicates** are 90%.
- Abundance of elements in the crust (weight %): **O 46.6, Si 27.7, Al 8.1, Fe 5.0, Ca 3.6, Na 2.8, K 2.6, Mg 2.1**.

1.1 Methods of Study of Minerals and Rocks

1. **Megascopic (hand specimen) study** - by naked eye / hand lens (10x): colour, luster, hardness, cleavage etc. - done in field and lab.
2. **Microscopic study** - thin section (0.03 mm) under **petrological (polarising) microscope** - optical properties, texture, mineral percentages.
3. **Chemical analysis** - wet analysis, atomic absorption, XRF.
4. **X-ray diffraction (XRD)** - identifies crystal structure; **SEM / EPMA** - micro-structure and composition.
5. **Blow-pipe analysis, thermal (DTA) & electron methods** - additional tests.

For Civil Engineers

- Megascopic study is the most useful: rocks and minerals are identified in the field and in the stone yard within minutes.

2. Physical Properties of Minerals (megascopic)

Property	Meaning / how to test
Colour	Appearance in reflected light; may vary with impurities (quartz: white, pink, purple). Use fresh surface.
Streak	Colour of mineral **powder** on unglazed porcelain plate. Hematite = cherry red; pyrite = greenish black.
Luster	Way light reflects: **metallic, sub-metallic, vitreous (glassy), resinous, pearly, silky, greasy, earthy, adamantine.**
Form / habit	Shape: prismatic, tabular, fibrous (asbestos), massive, granular, botryoidal, foliated, acicular.
Cleavage	Tendency to split along smooth flat planes: perfect, good, imperfect (mica - 1 direction, calcite - 3 directions not at 90°).
Fracture	Irregular break in non-cleavage direction: **conchoidal** (quartz), uneven, splintery, hackly, earthy.
Hardness	Resistance to scratching - **Mohs' scale** (1-10).
Specific gravity	Density relative to water: determined with Walker's steelyard / pycnometer. Metallic minerals have high SG.

Tenacity	Brittle, malleable, sectile, flexible, elastic (mica).
Other	Magnetism (magnetite), taste (halite), effervescence with dil. HCl (calcite), transparency, fluorescence, double refraction.

2.1 Mohs' Scale of Hardness

No.	Mineral	No.	Mineral	Field test
1	Talc	6	Orthoclase	Fingernail (2.5), copper coin (3.5)
2	Gypsum	7	Quartz	Knife blade / glass (5.5)
3	Calcite	8	Topaz	Steel file (6.5)
4	Fluorite	9	Corundum	Cuts glass
5	Apatite	10	Diamond	Hardest known

- Memory: **T**all **G**irls **C**an **F**latter **A**ll **O**ld **Q**uiet **T**all **C**ats **D**ear.

3. Common Rock-Forming Minerals

3.1 Feldspar Group (most abundant: ≈ 50-60% of crust)

- Alumino-silicates of **K, Na, Ca**. Two sub-groups:
  - **Orthoclase / Microcline (K-feldspar)** $KAlSi_3O_8$: colour white, pink, flesh; vitreous; **H = 6**; cleavage 2 directions ~90°; SG 2.56. Pink colour of granite.
  - **Plagioclase (Na-Ca)**: albite -> anorthite; white to grey; H = 6-6.5; cleavage 2 directions (86°); SG 2.6-2.76; shows **twin striations**.
- Uses: ceramics, glass, porcelain; weathers to **clay** (kaolin).

3.2 Quartz Group (SiO_2)

- **Varieties**: rock crystal (colourless), **milky** quartz, **rose** quartz (pink), **amethyst** (violet), smoky, citrine; cryptocrystalline: **chalcedony, agate, jasper, flint, chert**.
- Properties: vitreous luster; **H = 7**; SG 2.65; **no cleavage, conchoidal fracture**; hexagonal prism with pyramid; transparent to translucent.
- Uses: glass, abrasive, silicon chips, concrete aggregate; most resistant mineral to weathering -> **sand**.

3.3 Mica Group (sheet silicates)

- **Muscovite** (white mica) $KAl_2(AlSi_3O_{10})(OH)_2$: colourless / silvery; **H 2-2.5**; **perfect basal cleavage** -> thin, flexible, elastic sheets; SG 2.8-3.
- **Biotite** (black mica): $K(Mg,Fe)_3(AlSi_3O_{10})(OH)_2$: brown-black, pearly; same cleavage; SG 2.7-3.1.
- Uses: **electrical insulators**, paints, cosmetics; decays to clay and iron oxides.

Mineral	Colour / Luster	H	Cleavage / Form	SG	Remarks & uses
Olivine $(Mg,Fe)_2SiO_4$	Olive green; vitreous	6.5-7	None; conchoidal; granular	3.3	Peridot gem; ultra-basic rocks; weathers fast
Augite (pyroxene)	Dark green-black; vitreous	5-6	2 sets at ~87-93°; short prism	3.3	In basalt, gabbro; 8-sided cross section
Hornblende (amphibole)	Black-dark green; vitreous	5-6	2 sets at 56° & 124°; long prism	3.2	In granite, diorite, gneiss; 6-sided section
Asbestos (fibrous amphibole / serpentine)	White-grey-green; silky	5-6	Fibrous, flexible	2.5-3	Fire-proof, insulation; **health hazard**
Talc (hydrous Mg silicate)	White-pale green; pearly, greasy	1	Foliated, soapy feel	2.7	Talcum powder, soapstone, paint
Chlorite	Green; pearly	2-2.5	One perfect cleavage; flakes **flexible not elastic**	2.6-3	Low-grade metamorphic rocks (schist)
Kyanite Al_2SiO_5	Blue; vitreous-pearly	4-5 along / 6-7 across	One perfect + one good; bladed	3.6	Refractory; indicates **high-pressure metamorphism**
Garnet	Red-brown; vitreous-resinous	6.5-7.5	None; dodecahedral crystals	3.5-4.3	Gem, abrasive; in mica schist, gneiss

Calcite $CaCO_3$	White/colourless; vitreous	3	**Perfect 3 directions (rhombohedral)**	2.7	Fizzes with cold dil. HCl; cement, lime, marble

Quick identification clues

- Olivine = green, no cleavage. Hornblende vs Augite: 124° vs 87° cleavage angle. Kyanite = blue, two hardnesses. Calcite = effervescence. Talc = softest.

4. Ore Forming (Economic) Minerals

Mineral	Composition / Colour	H	Streak / Luster	SG	Uses / Occurrence
Pyrite "fool's gold"	FeS_2; brass yellow	6-6.5	Greenish-black; metallic	5	Sulphur, sulphuric acid; cubes striated
Hematite	Fe_2O_3; steel-grey to red	5.5-6.5	**Cherry red**; metallic	5.2	Principal **iron ore**; red pigment
Magnetite	Fe_3O_4; iron-black	5.5-6.5	Black; metallic	5.2	Rich iron ore; **strongly magnetic**
Galena	PbS; lead-grey	2.5	Lead-grey; metallic	7.5	Ore of **lead**; cubic cleavage
Pyrolusite	MnO_2; black-grey	2-2.5	Black; sub-metallic, soils fingers	4.8	Main **manganese** ore; steel, batteries
Graphite	C; black-steel grey	1-2	Black; greasy, metallic	2.2	Pencils, lubricant, electrodes; soft, greasy
Chromite	$FeCr_2O_4$; black	5.5	Brown; sub-metallic	4.5	Only ore of **chromium**; stainless steel
Bauxite	$Al_2O_3.nH_2O$; white-red-brown, earthy	1-3	White; earthy, pisolitic	2.5	Ore of **aluminium**; laterite product

- India: **iron** (Odisha, Jharkhand, Karnataka), **bauxite** (Odisha, A.P.-Araku), **manganese** (Odisha, Maharashtra), **chromite** (Sukinda, Odisha), **lead-zinc** (Rajasthan,

Zawar), **mica** (Nellore, A.P.).

5. Petrology - Study of Rocks

Rocks are of **three genetic types**; all are linked by the **rock cycle**.

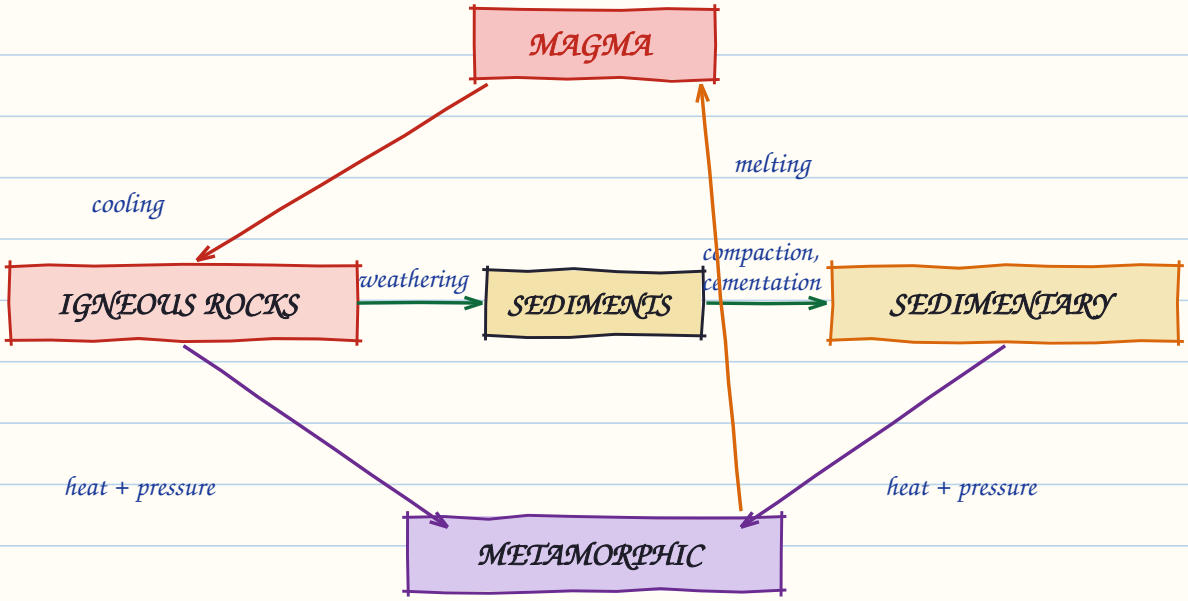


Fig 1: The Rock Cycle

5.1 Igneous Rocks (Primary / parent rocks)

- Formed by **cooling and solidification of magma (molten rock below surface) or lava (on surface)**. Make up ≈ 95% of the crust (volume).
- **Classification by place of consolidation:**
  - **Plutonic (abyssal)** - deep, very slow cooling, **coarse-grained** (granite, gabbro, diorite).
  - **Hypabyssal (dyke / sill)** - intermediate depth, medium grain, porphyritic (dolerite, pegmatite).
  - **Volcanic (extrusive)** - surface, rapid cooling, **fine-grained or glassy** (basalt, rhyolite, obsidian).
- **Classification by silica content:**

Type	SiO_2 %	Minerals	Examples
Acid	> 65	Quartz, K-feldspar, mica; light colour	Granite, rhyolite
Intermediate	55-65	Plagioclase, hornblende	Diorite, andesite

Basic	45-55	Ca-plagioclase, augite; dark	Gabbro, dolerite, basalt
Ultrabasic	< 45	Olivine, pyroxene; very dark	Peridotite, dunite

Textures of Igneous Rocks (size, shape, arrangement of grains)

- **Phaneritic** - grains visible to naked eye (granite); **Aphanitic** - too fine to see (basalt); **Glassy** - no crystals (obsidian).
- **Porphyritic** - large crystals (**phenocrysts**) in a fine **groundmass** (two stages of cooling).
- **Equigranular / granular** - grains of equal size; **Pegmatitic** - very large crystals (> 1 cm); **Ophitic** - laths of plagioclase enclosed in augite (dolerite).
- **Vesicular** - gas holes; **Amygdaloidal** - holes filled with calcite/quartz.

Structures of Igneous Rocks

- **Vesicular, amygdaloidal, ropy / pahoehoe** and **pillow** (underwater lava), **flow banding**, **columnar jointing** (basalt), **sheet joints, xenoliths**.

Forms of Igneous Bodies

(dyke cuts bedding; sill is parallel to bedding)

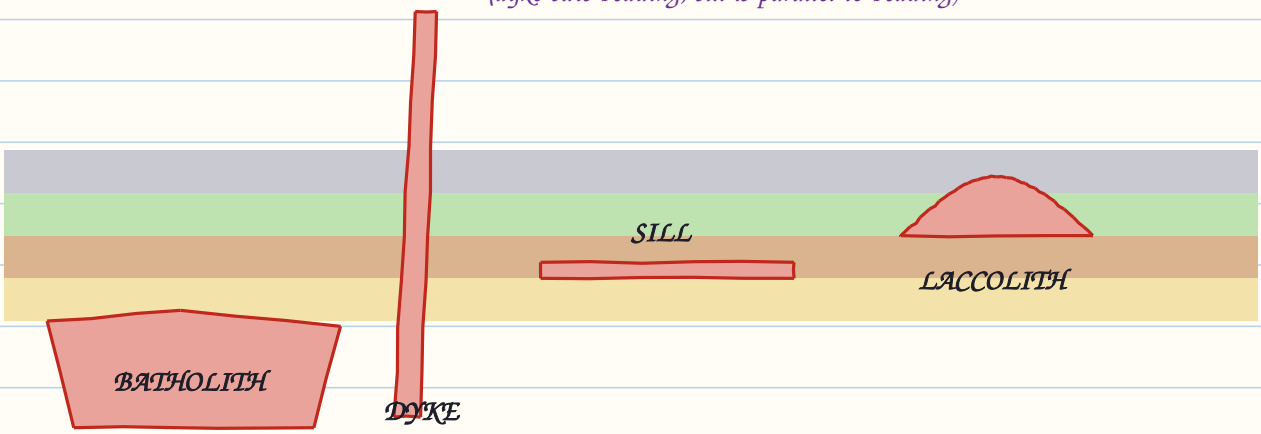


Fig 2: Forms of igneous intrusions

- **Intrusive:** batholith (largest, granite), **stock, laccolith** (lens-shaped dome), **lopolith** (saucer), **sill** (sheet parallel to beds), **dyke** (sheet cutting across beds), **volcanic neck**.
- **Extrusive:** lava flows, **lava plateau (Deccan traps)**, volcanic cones, pyroclastics (ash, bombs).

5.2 Megascopic Study of Igneous Rocks

Rock	Megascopic description	Engineering use

Granite - pink	Coarse, **pink orthoclase** + quartz + biotite; hard, dense, SG 2.6-2.7, compressive strength 100-250 MPa	Dimension stone, monuments, **dam & bridge foundations**, aggregate
Granite - grey	Grey plagioclase / quartz + biotite and hornblende; uniform	Building stone, paving, **polished slabs**
Granite - green	Greenish feldspar (amazonite / altered, chloritised)	Ornamental / cladding
Pegmatite	**Very coarse** crystals (cm to m) of quartz, feldspar, mica; occurs as veins	Source of mica, feldspar, gems; **weak zones** for foundations
Dolerite	Dark, medium-grained, **ophitic**; plagioclase + augite; SG 2.9	**Excellent road metal**, ballast, concrete aggregate; occurs as dykes
Basalt	Black-dark grey, **fine-grained**, may be vesicular or amygdaloidal; columnar joints; SG 2.9-3.0	Aggregate, road metal, **dams (Deccan)**; weathers to black-cotton soil

5.3 Sedimentary Rocks (Secondary rocks)

- Formed by **weathering -> erosion -> transport -> deposition -> lithification** (compaction + cementation) of older rocks, chemical precipitation or organic remains. Cover ≈ 75% of land surface but only ≈ 5% of crust volume.
- **Classification:**

Class	Origin	Examples
Mechanical / clastic (detrital)	Fragments of rocks	Conglomerate, **sandstone, shale**, breccia
Chemical	Precipitation from solution	**Limestone**, rock salt, gypsum
Organic	Remains of animals / plants	Coal, shell limestone, chalk
Residual	In-situ weathering residue	**Laterite**, bauxite

- Grain size (Wentworth): boulder > cobble > pebble (> 2 mm) -> **sand (2-1/16 mm)** -> **silt (1/16-1/256)** -> **clay (< 1/256 mm)**.

Structures (primary) of sedimentary rocks

- **Stratification / bedding** (layers, planes of deposition), **lamination** (< 1 cm), **cross-bedding** (inclined layers in dunes/rivers), **graded bedding**, **ripple marks**, **mud**

cracks, rain prints, fossils, concretions, nodules.

- Texture: clastic (rounded / angular grains), crystalline (chemical), bioclastic; cement = silica, calcite, iron oxide.

5.4 Megascopic Study of Sedimentary Rocks

Rock	Megascopic description	Engineering use
Shale	Very fine, laminated, splits into thin sheets (fissile); grey-black-red; soft, absorbs water, slakes	Raw material for bricks, cement; poor foundation / unstable slope
Sandstone	Gritty feel; sand grains of quartz cemented by silica / calcite / iron oxide; cross-bedding; colour white, buff, red	Building and paving stone, aggregate; silica-cemented = strong, ferruginous weaker
Limestone	Calcite; grey-white; fine to crystalline; effervescence with dil. HCl; may have fossils; H = 3	Cement, lime, flux, building stone; solution cavities (karst) - leakage in reservoirs
Laterite	Reddish-brown porous rock, rich in Fe and Al oxides; soft when quarried, hardens on exposure; pitted surface	Building blocks (Kerala, Konkan), road metal; bauxite in some

5.5 Metamorphic Rocks (Transformed rocks)

- Metamorphism = change of mineral composition / texture / structure of pre-existing rocks in the solid state due to heat, pressure and chemically active fluids, without melting.
- Types: Contact (thermal) - near igneous intrusion, heat only (marble, hornfels); Regional (dynamothermal) - large areas, heat + pressure (schist, gneiss); Dynamic (cataclastic) - along faults, pressure (mylonite); Hydrothermal / metasomatism.
- Grades: low (slate) -> medium (phyllite, schist) -> high (gneiss, granulite).
- Index minerals: chlorite -> biotite -> garnet -> kyanite -> sillimanite (increasing grade).

Textures and Structures

- Foliated (layered): slaty cleavage, schistosity, gneissose banding. Non-foliated: granulose, granoblastic (marble, quartzite), hornfelsic.

- **Porphyroblastic** (large new crystals in fine matrix) - garnet schist.
- Structures: **slaty cleavage, schistosity, gneissic banding, lineation, augen.**

Foliation order (increasing grade)

- Slate (very fine, rings when struck) -> Phyllite (silky sheen) -> Schist (visible flaky mica, wavy) -> Gneiss (coarse light and dark bands)

5.6 Megascopic Study of Metamorphic Rocks

Rock	Parent rock	Megascopic description	Engineering use
Quartzite	Sandstone	Very hard, **interlocking quartz** grains, glassy, no foliation, conchoidal fracture; white-pink	Road metal, aggregate, railway ballast; **strong dam foundations**
Gneiss	Granite / sediments	Coarse, **banded** - light (quartz-feldspar) and dark (mica, hornblende) layers	Building stone, paving; **good foundations** (planes of weakness along banding)
Schist	Shale / basalt	Medium-coarse, **wavy flaky** minerals (mica, chlorite, talc, garnet); splits readily	**Poor**: weak, slippery along schistosity - slope / tunnel problems
Marble	Limestone / dolomite	Crystalline, saccharoidal, **effervescence with HCl**, white or coloured veins; H = 3-4	Monuments, flooring, statues, decorative stone
Khondalite	Aluminous (pelitic) sediments	Granulite-grade, **garnet + sillimanite + quartz + graphite**, coarse banded; abundant in **Eastern Ghats (Andhra, Odisha)**	Local building stone; weathers to red soil
Slate	Shale / clay	**Very fine**, perfect slaty cleavage - splits into thin flat slabs; grey-black; rings when struck	Roofing, flooring, blackboards, damp-proof course

5.6A Additional Notes - Origin and Field Identification

Silicate mineral structure (why silicates dominate)

- Basic unit = **SiO_4 tetrahedron**. Linking pattern gives groups: **isolated** (olivine, garnet) -> **single chain** (pyroxene / augite) -> **double chain** (amphibole / hornblende) -> **sheet**

(mica, talc, chlorite) -> **framework** (quartz, feldspar).

- More linking = **harder, more stable** mineral: that is why quartz is the last to weather.
- **Bowen's reaction series:** first to crystallise (olivine, Ca-plagioclase) -> last (muscovite, quartz). **Discontinuous** (Fe-Mg minerals) and **continuous** (plagioclase) branches.

Formation of Laterite and Bauxite

- **Laterite**: intense **chemical weathering** of basalt / granite / khondalite in **hot, humid, alternating wet-dry climate** - silica and bases are leached out; **Fe and Al oxides** remain. Found in **Kerala, Karnataka coast, Odisha, East Godavari (A.P.)**.
- **Bauxite**: laterite rich in **alumina**, mined for aluminium (Araku Valley, Visakhapatnam).

Common Field Tests

- **HCl test:** calcite / limestone / marble fizz; **Scratch test:** knife (5.5), nail (2.5); **Streak test:** hematite red.
- **Magnet test:** magnetite; **Lustre:** pyrite metallic; **Grain size:** hand lens.
- **Colour index** of igneous rocks = % dark minerals: **leucocratic** (light), **mesocratic**, **melanocratic** (dark).

Rock Identification Flow

Question	If YES	If NO
Is it layered / banded / flaky?	Metamorphic (slate, schist, gneiss) or sedimentary (shale)	Go to next
Are crystals interlocking and visible?	Igneous coarse (granite, gabbro) or marble / quartzite	Go to next
Is it very fine, dark, vesicular?	Basalt (igneous volcanic)	Clastic or chemical rock
Is it made of rounded / angular grains with cement?	Sandstone / conglomerate	Limestone / laterite

- **Colour clues:** pink granite -> K-feldspar; grey granite -> plagioclase; green granite -> chloritised or amazonite feldspar; **black** -> basalt / dolerite; **red-brown, porous** -> laterite; **white, fizzing** -> limestone / marble.

5.7 Engineering Properties - Quick Comparison

- **Igneous (granite, basalt, dolerite):** strong, massive, low porosity - best for foundations, aggregate, dam sites. Watch for **joints, dykes, weathered zones**.
- **Sedimentary:** variable. **Sandstone** good if well cemented; **shale** weak and swells; **limestone** has cavities and solution channels.
- **Metamorphic:** quartzite, gneiss, marble strong; **schist, slate, phyllite** weak along foliation planes - failures in cuts and tunnels.

Rock	Compressive strength (MPa)	Porosity (%)	Remarks
Granite	100-250	< 1	Best foundation rock
Basalt	100-300	0.1-1	Sound; fractured zones leak
Sandstone	20-170	5-25	Variable by cement
Limestone	30-250	1-15	Solution cavities
Shale	5-100	5-30	Weak; slakes
Quartzite	150-300	< 1	Very strong
Marble	50-150	< 2	Good, moderate
Slate	50-180	< 1	Anisotropic

Exam checklist

- Write definition + classification table + 3 textures/structures + engineering uses. Draw neat sketches of the rock cycle and igneous forms (dyke, sill, laccolith, batholith).

5.8 Short Revision Questions

1. **Why is quartz found as sand?** It is hard, chemically stable and resists weathering.
2. **Mineral with cleavage in 3 directions not at 90°?** Calcite (rhombohedral cleavage).
3. **Mineral with H = 1 and soapy feel?** Talc.
4. **Name the ore of aluminium, iron, lead, manganese and chromium.** Bauxite, hematite / magnetite, galena, pyrolusite, chromite.
5. **Which rock is a good road metal?** Dolerite, basalt, quartzite, granite.
6. **Define porphyritic texture.** Large phenocrysts in a fine groundmass.
7. **What are the parent rocks of marble, quartzite and slate?** Limestone, sandstone,

shale.